Electromagnetism Part 7: Dielectrics and Electric Displacement

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1 Introduction

Materials can be roughly divided into conductors and insulators. In a conductor there are charges that are free to move, while in an insulator all charges are bound. In a specific type of insulator, called **dielectrics**, charges are able to move a very small distance, typically within an atom or a molecule. This movement happens in response to an electric field, resulting in polarisation. This is what we will study here.

2 Polarisation of Dielectrics

2.1 Induced Dipoles

While atoms are electrically neutral overall, the nucleus is positively charged while the electrons are negatively charged. When an electric field is applied, these two parts tend to be pushed apart. This is counteracted by the attractive force between the nucleus and the electron cloud, such that an equilibrium is reached. The atom is now polarised and has acquired a dipole moment $\vec{p}$. This is approximately proportional to, and points in the same direction as, the applied field $\vec{E}$, i.e:

$$\vec{p} = \alpha \epsilon_0 \vec{E}$$

where α is a constant that depends on the particular type of atom.

2.2 Polar Molecules

Some molecules have a dipole moment even when not in the presence of an electric field due to the way the electrons end up being distributed within the molecule. These are called **polar molecules**. Water is a good example.

When polar molecules are placed in an electric field, they (like all dipoles) experience a torque,

$$\tau = \vec{p} \times \vec{E}$$

which tends to rotate the molecule to align the dipole moment with the field. Once $\vec{p}$ and $\vec{E}$ are aligned, the cross product is zero, and the torque becomes zero.

2.3 Macroscopic Polarisation

We have seen what happens to atoms or polar molecules in the presence of electric fields. Therefore, any macroscopic dielectric material placed in an electric field will end up having a very large number of dipoles aligned with the field. We define the polarisation, $\vec{P}$, as the **dipole moment per unit volume**.

Polarisation is illustrated in Figure 1.

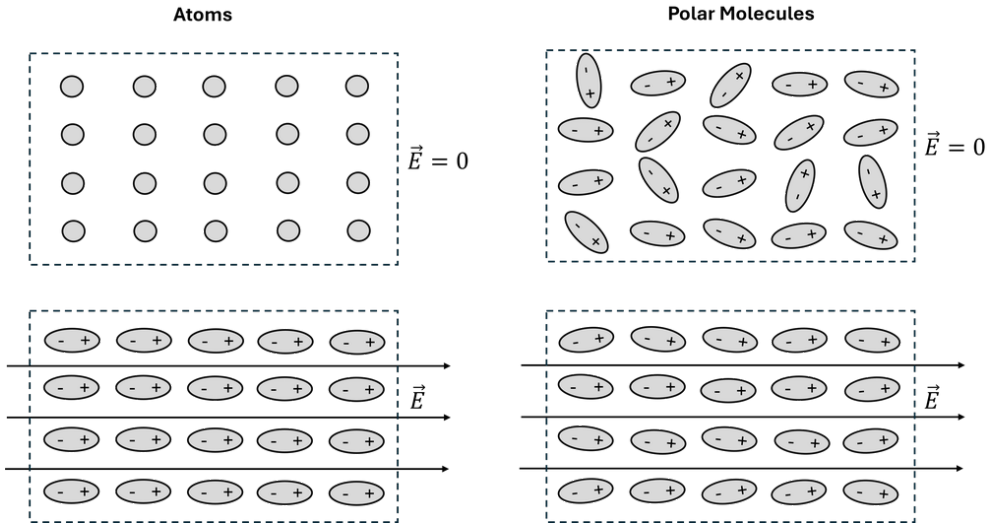


Figure 1: Polarisation of atoms and polar molecules.

3 Field of a Polarised Material

If a material is polarised by an external field, it will now generate its own electric field. We know the field from a single dipole, so we can work out the total field by integrating over all the dipoles. This is easier if we work with potential. For a single dipole at $\vec{r'}$, the potential at position $\vec{r}$ is,

$$\phi_{\rm dip}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\frac{\vec{p}\cdot(\vec{r}-\vec{r'})}{|\vec{r}-\vec{r'}|^3} \tag{1}$$

A nice maths trick is that we can rewrite this using div (check what $\nabla(1/\vec{r})$ is!),

$$\phi_{\rm dip}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\,\vec{p}\cdot\nabla'\left(\frac{1}{|\vec{r}-\vec{r'}|}\right) \tag{2}$$

where we have used the ∇' notation to make it clear that the operator acts on r' not r.

Now, for a polarised dielectric, the dipole moment per unit volume is the polarisation vector $\vec{P}(\vec{r'})$, so that an infinitesimal volume element $d\tau'$ has dipole moment,

$$d\vec{p} = \vec{P}(\vec{r'})\,d\tau'.$$

The total potential is obtained by integrating over the volume of the dielectric,

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\int \vec{P}(\vec{r'})\cdot\nabla'\left(\frac{1}{|\vec{r}-\vec{r'}|}\right)d\tau'. \tag{3}$$

Using the vector identity,

$$\vec{P}\cdot\nabla' f = \nabla'\cdot(\vec{P}f) - f\,(\nabla'\cdot\vec{P}),$$

with $f = |\vec{r}-\vec{r'}|^{-1}$, we obtain,

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left[\int \nabla'\cdot\left(\frac{\vec{P}(\vec{r'})}{|\vec{r}-\vec{r'}|}\right)d\tau' - \int\frac{\nabla'\cdot\vec{P}(\vec{r'})}{|\vec{r}-\vec{r'}|}d\tau'\right]. \tag{4}$$

Applying the divergence theorem to the first term gives,

$$\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left[\oint\frac{\vec{P}(\vec{r'})\cdot\hat{\vec{n}}'}{|\vec{r}-\vec{r'}|}\,da' - \int\frac{\nabla'\cdot\vec{P}(\vec{r'})}{|\vec{r}-\vec{r'}|}\,d\tau'\right]. \tag{5}$$

Defining the bound surface and volume charge densities as,

$$\sigma_b = \vec{P}\cdot\hat{\vec{n}}, \qquad \rho_b = -\nabla\cdot\vec{P},$$

the potential can be written as

$$\boxed{\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\left[\oint\frac{\sigma_b(\vec{r'})}{|\vec{r}-\vec{r'}|}\,da' + \int\frac{\rho_b(\vec{r'})}{|\vec{r}-\vec{r'}|}\,d\tau'\right]}. \tag{6}$$

The potential is therefore the sum of the potential produced by a volume charge density $\rho_b = -\nabla \cdot \vec{P}$ and a surface charge density $\sigma_b = \vec{P} \cdot \hat{n}$.

Why is this the case? Consider what happens when we polarise a thin tube of the material, as shown in Figure 2. We polarise each of the atoms or molecules, with all the dipoles aligned. The tail of each dipole cancels the head of the previous dipole. The only place this doesn't happen is at the ends of the tube, the surface boundaries, where we run out of dipoles to cancel with. So we end up with a surface charge, q.

Remember that the polarisation is the dipole moment per unit volume. So in a boundary region of thickness d, equal to the length of the final dipole, we end up with a dipole moment of magnitude $p = PAd$. But for a dipole, $p = qd$ and so charge $q = PA$, and so the surface charge density is $\rho_b = q/A = P$.

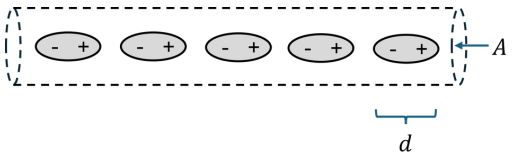


Figure 2: Thin tube of a polarised dielectric.

That only works if the boundary surface is perpendicular to the 'tube', or more generally, $\vec{P}$, otherwise the surface charge density is $\vec{P} \cdot \hat{n}$.

What about the other part, the volume charge density? This is coming from the divergence of $\vec{P}$. So this is non-zero when there is some variation of $\vec{P}$ over the volume.

Note that in the above we were implicitly talking about the macroscopic potential (and field). If we zoom in far enough then there would be a much more complex structure depending on exactly where we are in relation to a dipole. But we generally don't care about that, we are only interested in the average potential (or field) over some area that is large compared to the atomic separation.

4 Electric Displacement

Now we know we are getting accumulations of bound charge in a polarised dielectric, we can think about the field produced by this charge (not to be confused with the field causing the polarisation in the first place).

To be general, we also include any free charge that might be in the dielectric (e.g. some ions that happen to be in the dielectric), so that the total charge density is the sum of bound and free charge densities,

$$\rho = \rho_b + \rho_f$$

Now, Gauss's law ($\nabla \cdot \vec{E} = \rho/\epsilon_0$) becomes,

$$\epsilon_0 \nabla \cdot \vec{E} = \rho_b + \rho_f$$

,

and since we had earlier had $\nabla \cdot \vec{P} = -\rho_b$, then,

$$\nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_f$$

we define the displacement, $\vec{D}$ as,

$$\boxed{\vec{D} = \epsilon_0 \vec{E} + \vec{P}}$$

and so we have a new way to write Gauss's Law,

$$\boxed{\nabla \cdot \vec{D} = \rho_f,}$$

or in integral form,

$$\boxed{\oint \vec{D} \cdot d\vec{a} = Q_{f,\text{enc}},}$$

where $Q_{f,enc}$ is the free charge enclosed in the volume.

Note 1: The key difference between $\vec{E}$ and $\vec{D}$ (other than the factor of ϵ_0: $\vec{D}$ depends only on the free charge, while $\vec{E}$ also depends on the bound charge created by polarisation.

Note 2: In general we know the free charge, but it's hard to figure out the bound charge, so we can figure out $\vec{D}$ rather than $\vec{E}$. In the previous lectures we were not considering dielectrics or polarisation, we always had $\vec{P} = 0$ and so $\epsilon_0 \vec{E} = \vec{D}$ in any case.

Note 3: While we have $\nabla \times \vec{E} = 0$ always, this is **not** true for $\vec{D}$. In fact for electrostatic it is equal to $\nabla \times \vec{P}$, which in general is not zero. So, there is no equivalent of the scalar potential (V) for $\vec{D}$.

4.1 Example of insulated wire

We have a straight wire (i.e. a conductor) with uniform linear charge density λ and radius a, as shown in Figure 3. The wire is insulated with a dielectric with outer radius b. What is the electric displacement, $\vec{D}$?

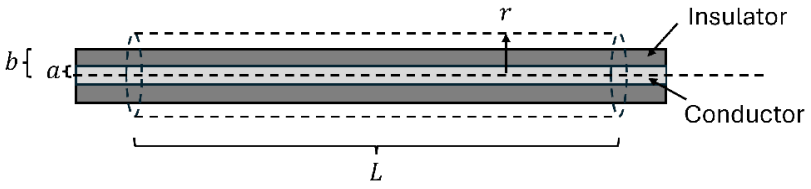


Figure 3: Insulated wire example.

We know how to do this outside the wire insulation, just apply Gauss's Law, but now using $\vec{D}$ rather than $\vec{E}$. We choose a cylindrical Gaussian surface with radius, r and length L. The field is always parallel to the two ends of the cylinder, so $\vec{D} \cdot d\vec{a}$ is zero there, and it is perpendicular to the curved surface, and constant in magnitude. We therefore have,

$$\oint \vec{D} \cdot d\vec{a} = Q_{f,enc} \tag{7}$$

$$D(2\pi r L) = \lambda L \tag{8}$$

$$D = \frac{\lambda}{2\pi r} \tag{9}$$

Since the electric displacement points outwards,

$$\vec{D} = \frac{\lambda}{2\pi r}\hat{r}$$

This is not a big surprise, it's the same as $\epsilon_0 \vec{E}$ when we don't have a dielectric surrounding the wire. Of course now $\vec{E}$ would be something else due to the polarisation of the dielectric. In fact it's still true that,

$$\vec{E} = \frac{\lambda}{2\pi r \epsilon_0}\hat{r}$$

when we are outside the dielectric coating, since there $\vec{P} = 0$, and assuming that the free charge on the conductor is not changed by the bound charge on the dielectric. Inside the dielectric we don't know $\vec{E}$.

5 Linear Dielectrics

The polarisation of dielectrics is approximately linear with electric field, at least for weak electric fields. We write this as,

$$\vec{P} = \epsilon_0 \chi_e \vec{E}$$

where χ_e is the electric susceptibility of the particular material. Note that χ_e is a dimensionless quantity.

Unfortunately, this is not as simple as it looks. The field $\vec{E}$ is not just the applied, external field, it also includes the field generated by the polarisation. But this then changes the polarisation, which changes the field again, and so on. Of course this eventually settles down somewhere, as the two fields are in opposition, they don't reinforce each other, but it's not always so easy to figure out where.

In linear media we also have,

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = \epsilon_0 \chi_e \vec{E} = \epsilon_0 (1 + \chi_e) \vec{E}$$

$\vec{D}$ is therefore proportional to $\vec{E}$. The proportionality constant we define as ϵ, the **permittivity** of the dielectric material, with,

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

Finally, we also define the **relative permittivity** or **dielectric constant**, ϵ_r, which is,

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 1 + \chi_e$$

where $\epsilon_r > 1$ for normal dielectrics.

6 Dielectric Capacitors

Finally, we come to an important application of dielectrics. Recall that capacitors store energy. The simplest capacitor to think about is a pair of parallel conducting plates. When a potential difference is applied across the plates, charge flows from one to the other. The amount of charge, Q, that is transferred depends on the capacitance, C, of the capacitor, which depends on its geometry (e.g. size of plates) and the potential difference, such that,

$$Q = CV.$$

We do work 'charging' the capacitor, and this energy is then released if the potential difference source is disconnected, and the charge is allowed to flow back to where it started. The energy stored when the capacitor is fully charged is $E = (1/2)CV^2$. So, if we want to store more energy, we would like to increase the capacitance.

What happens if we put a dielectric between the plates? Now we expect to be able to transfer more charge, and store more energy for a given V. This is because the dielectric will become polarised, creating its own field and bound charge which works against the charge that had already flowed to the plates. We store more energy by a factor of ϵ_r, and so we say the capacitance has increased compared to the capacitance if it was filled with vacuum by that amount, and,

$$C = \epsilon_r C_{\text{vac}}$$

where C_{vac} is the capacitance with no dielectric. This is why we have 'dielectric capacitors'. The downside is that if an excessive voltage is applied, or if the dielectric degrades due to age or bad treatment, the dielectric breaks down and allows current to flow between the plates, leading to a pop, a nasty smell, and the failure of your electronics.